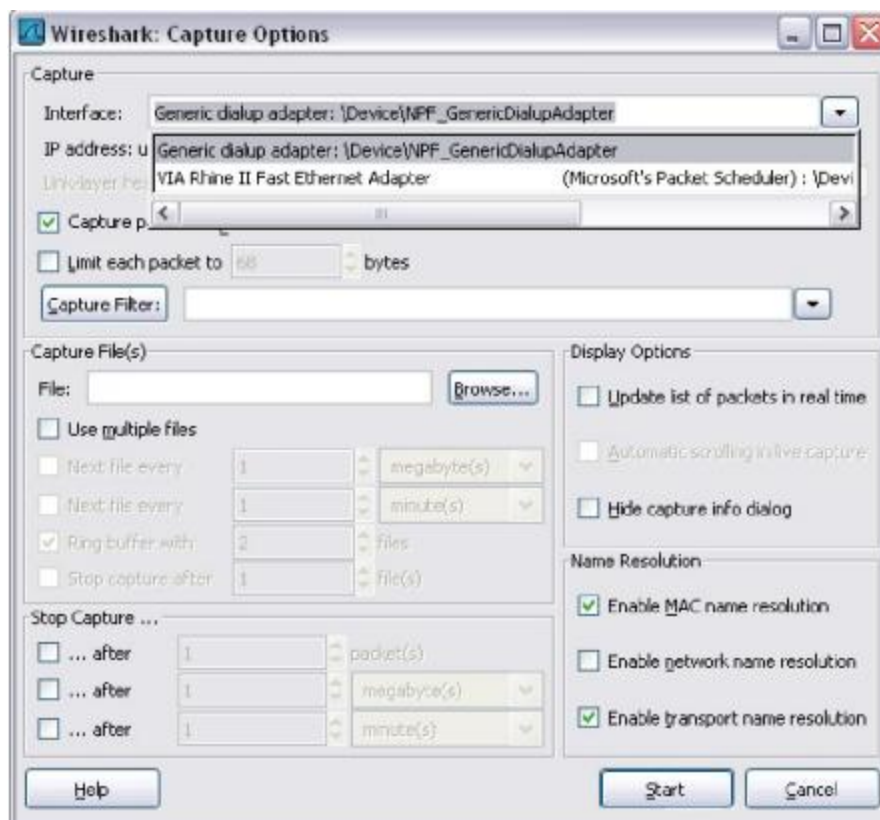
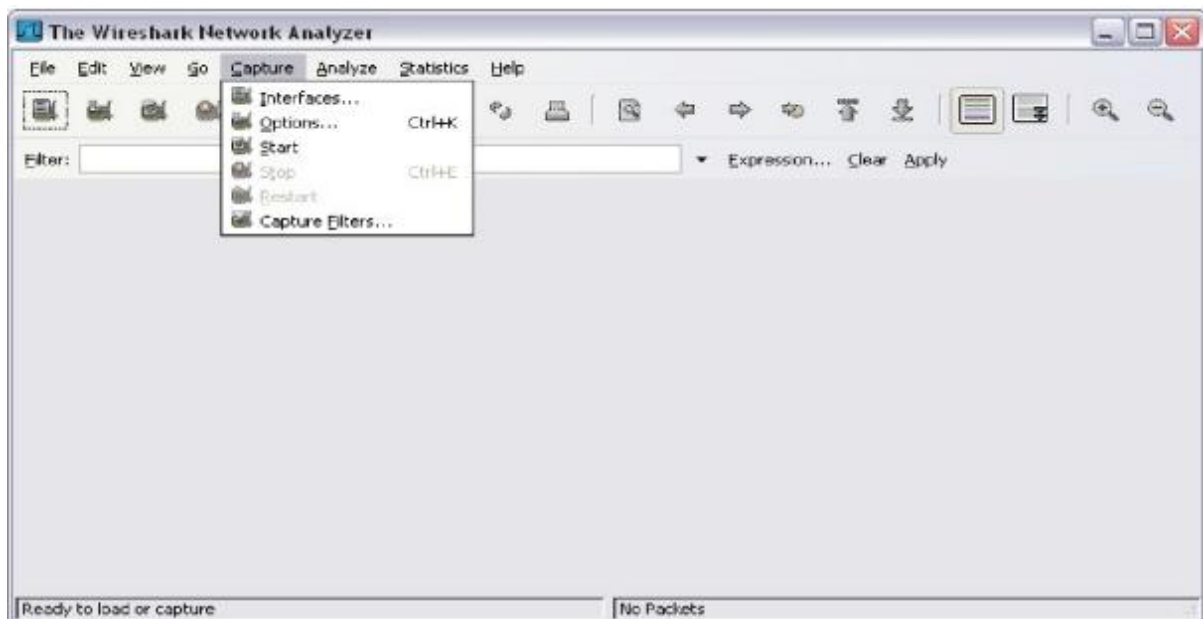


Use of the Wireshark tool to capture and examine packets



Wireshark: Capture Options

Capture

Interface: VIA Rhine II Fast Ethernet Adapter (Microsoft's Packet Scheduler) : \

IP address: 192.168.0.6

Link-layer header type: Ethernet Buffer size: 1 megabyte(s) Wireless Settings

☒ Capture packets in promiscuous mode

☐ Limit each packet to 50 bytes

Capture Filter:

Capture File(s)

File: Browse...

☐ Use multiple files

☐ Next file every: 1 megabyte(s)

☐ Next file every: 1 minute(s)

☒ Ring buffer with 2 files

☐ Stop capture after 1 file(s)

Stop Capture ...

☐ ... after 1 packet(s)

☐ ... after 1 megabyte(s)

☐ ... after 1 minute(s)

Display Options

☐ Update list of packets in real time

☐ Automatic scrolling in live capture

☐ Hide capture info dialog

Name Resolution

☒ Enable MAC name resolution

☐ Enable network name resolution

☒ Enable transport name resolution

Help Start Cancel

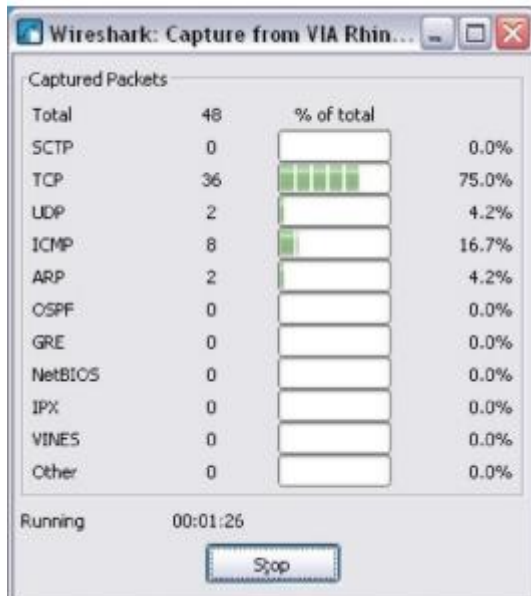
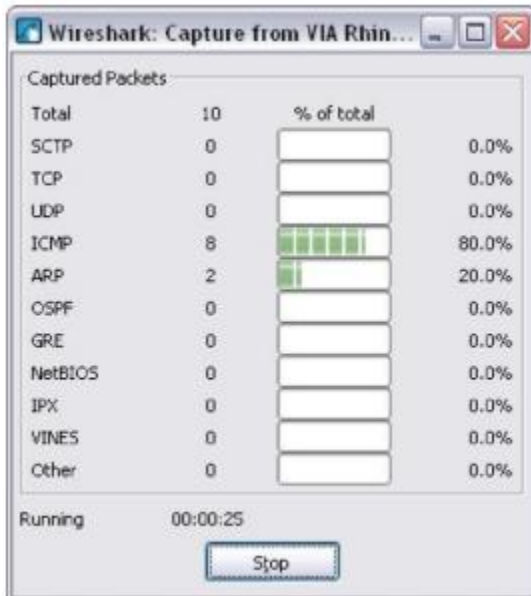
Wireshark: Capture from Generic ...

Captured Packets

	Total	% of total
SCTP	0	0.0%
TCP	0	0.0%
UDP	0	0.0%
ICMP	0	0.0%
ARP	0	0.0%
OSPF	0	0.0%
GRE	0	0.0%
NetBIOS	0	0.0%
IPX	0	0.0%
VINES	0	0.0%
Other	0	0.0%

Running 00:00:05

Stop



Wireshark interface showing a packet capture. The main pane displays a list of packets with columns: No., Time, Source, Destination, Protocol, and Info. A red box labeled "Packet List Pane" highlights the list. Below the list, the "Packet Details Pane" shows the structure of the selected packet (Frame 1, Ethernet II, Internet Protocol, Internet Control Message Protocol). A red box labeled "Packets Bytes Pane" highlights the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Info
1	0.000000	192.168.0.6	192.168.0.1	ICMP	Echo (ping) request
2	0.000074	192.168.0.1	192.168.0.6	ICMP	Echo (ping) reply
3	0.001524	0-LINK_92:7d:67	ASUSTEK_7c:35:4b	ARP	who has 192.168.0.6? Tell 192.168.0.1
4	0.001535	ASUSTEK_7c:35:4b	0-LINK_92:7d:67	ARP	192.168.0.6 is at 00:17:31:7c:35:4b
5	0.988933	192.168.0.6	192.168.0.1	ICMP	Echo (ping) request
6	0.989775	192.168.0.1	192.168.0.6	ICMP	Echo (ping) reply
7	1.988904	192.168.0.6	192.168.0.1	ICMP	Echo (ping) request
8	1.989724	192.168.0.1	192.168.0.6	ICMP	Echo (ping) reply
9	2.988683	192.168.0.6	192.168.0.1	ICMP	Echo (ping) request
10	2.989722	192.168.0.1	192.168.0.6	ICMP	Echo (ping) reply
11	60.355810	192.168.0.1	192.168.0.6	ICMP	Echo (ping) request
12	61.174087	208.0.178.0	192.168.0.6	ICMP	Echo (ping) reply
13	61.175108	192.168.0.6	www.wireshark.org	TCP	3471 > http [SYN] Seq=0 Len=0 MSS=1260
14	61.410076	www.wireshark.org	192.168.0.6	TCP	http > 3471 [SYN, ACK] Seq=0 Ack=1 Win=73
15	61.410126	192.168.0.6	www.wireshark.org	TCP	3471 > http [ACK] Seq=1 Ack=1 Win=64512 Len=0
16	61.410461	192.168.0.6	www.wireshark.org	HTTP	GET / HTTP/1.1
17	61.668353	www.wireshark.org	192.168.0.6	TCP	Chunk segment of a reassembled PDU
18	61.676122	www.wireshark.org	192.168.0.6	TCP	[TCP segment of a reassembled PDU]
19	61.676154	192.168.0.6	www.wireshark.org	TCP	3471 > http [ACK] Seq=447 Ack=2521 Win=645
20	61.919318	www.wireshark.org	192.168.0.6	TCP	[TCP segment of a reassembled PDU]

Packet Details Pane:

- Frame 1 (74 bytes on wire) (Captured on 0)
- Ethernet II, Src: Asustek_00:17:31:7c:35:4b, Dst: 192.168.0.1
- Internet Protocol Version 4, Src: 192.168.0.6, Dst: 192.168.0.1
- Internet Control Message Protocol, Type: Echo (ping) request

Packets Bytes Pane:

```

0000  00 50 ba 92 7d 67 00 17 31 7c 35 4b 00 00 00 00  .P...g...11K..E.
0010  00 3c 82 a8 00 00 80 01 00 00 00 00 00 00 00 00  .....
0020  00 01 98 00 38 5c c2 03 00 00 00 00 00 00 00  .....
0030  67 68 68 68 68 68 68 68 68 68 68 68 68 68 68 68  .....
0040  77 60 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f  .....
  
```

No.	Time	Source	Destination	Protocol	Info
1	0.000000	Cisco_9f:6c:c9	Spanning-tree-(for STP	STP	Conf. Root = 32769/00:0f:f7:9f:6c:c0 Cost =
2	2.000032	Cisco_9f:6c:c9	Spanning-tree-(for STP	STP	Conf. Root = 32769/00:0f:f7:9f:6c:c0 Cost =
3	4.000059	Cisco_9f:6c:c9	Spanning-tree-(for STP	STP	Conf. Root = 32769/00:0f:f7:9f:6c:c0 Cost =
4	4.072858	QuantaCo_bd:0c:7c	Broadcast	ARP	who has 10.1.1.254? Tell 10.1.1.1
5	4.073609	Cisco_cf:66:40	QuantaCo_bd:0c:7c	ARP	10.1.1.254 is at 00:0c:85:cf:66:40
6	4.073626	10.1.1.1	192.168.254.254	ICMP	Echo (ping) request
7	4.074122	192.168.254.254	10.1.1.1	ICMP	Echo (ping) reply
8	5.067535	10.1.1.1	192.168.254.254	ICMP	Echo (ping) request
9	5.068007	192.168.254.254	10.1.1.1	ICMP	Echo (ping) reply
10	6.000113	Cisco_9f:6c:c9	Spanning-tree-(for STP	STP	Conf. Root = 32769/00:0f:f7:9f:6c:c0 Cost =
11	6.067548	10.1.1.1	192.168.254.254	ICMP	Echo (ping) request
12	6.068019	192.168.254.254	10.1.1.1	ICMP	Echo (ping) reply
13	6.084103	Cisco_9f:6c:c9	Cisco_9f:6c:c9	LOOP	Reply
14	7.067603	10.1.1.1	192.168.254.254	ICMP	Echo (ping) request
15	7.068131	192.168.254.254	10.1.1.1	ICMP	Echo (ping) reply
16	8.000126	Cisco_9f:6c:c9	Spanning-tree-(for STP	STP	Conf. Root = 32769/00:0f:f7:9f:6c:c0 Cost =
17	9.975700	Cisco_9f:6c:c9	CDP/VTP/DTP/PagP/U	DTP	Dynamic Trunking Protocol
18	10.000134	Cisco_9f:6c:c9	Spanning-tree-(for STP	STP	Conf. Root = 32769/00:0f:f7:9f:6c:c0 Cost =

```

Frame 6 (74 bytes on wire (74 bytes captured)
Arrival Time: Jan 10, 2007 01:54:07.860436000
[Time delta from previous packet: 0.000017000 seconds]
[Time since reference or first frame: 4.073626000 seconds]
Frame Number: 6
Packet Length: 74 bytes
Capture Length: 74 bytes
[Frame is marked: False]
[Protocols in frame: eth:ip:icmp:data]
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp]
Ethernet II, Src: Quantaco_bd:0c:7c (00:c0:9f:bd:0c:7c), Dst: Cisco_cf:66:40 (00:0c:85:cf:66:40)
Destination: Cisco_cf:66:40 (00:0c:85:cf:66:40)
Source: Quantaco_bd:0c:7c (00:c0:9f:bd:0c:7c)
Type: IP (0x0800)
Internet Protocol, Src: 10.1.1.1 (10.1.1.1), Dst: 192.168.254.254 (192.168.254.254)
Version: 4
Header length: 20 bytes
Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00)
Total Length: 60
Identification: 0x0bf7 (3063)
Flags: 0x00
Fragment offset: 0
Time to live: 128
Protocol: ICMP (0x01)
Header checksum: 0x6421 [correct]
Source: 10.1.1.1 (10.1.1.1)
Destination: 192.168.254.254 (192.168.254.254)
Internet Control Message Protocol
Type: 8 (Echo (ping) request)
Code: 0
Checksum: 0x2a1c [correct]
Identifier: 0x0300
Sequence number: 0x2000

```

```

C:\Users\singh>tracert 8.8.8.8
Tracing route to google-public-dns-a.google.com [8.8.8.8]
over a maximum of 30 hops:
  0  <1 ms    <1 ms    <1 ms    192.168.1.1
  1  13 ms    20 ms    15 ms    120.57.48.1
  2  14 ms    13 ms    13 ms    triband-del-59.180.212.202.bol.net.in [59.180.212.202]
  3  14 ms    14 ms    14 ms    triband-del-59.180.210.150.bol.net.in [59.180.210.150]
  4  14 ms    13 ms    13 ms    125.20.37.21
  5  14 ms    16 ms    14 ms    182.79.181.230
  6  60 ms    59 ms    60 ms    182.79.190.57
  7  67 ms    101 ms   92 ms    182.79.198.162
  8  63 ms    63 ms    62 ms    72.14.197.166
  9  55 ms    55 ms    54 ms    108.170.253.121
 10 122 ms    89 ms    88 ms    216.239.63.213
 11 87 ms    86 ms    86 ms    216.239.47.109
 12 *        *        *        Request timed out.
 13 *        *        *        Request timed out.
 14 *        *        *        Request timed out.
 15 *        *        *        Request timed out.
 16 *        *        *        Request timed out.
 17 *        *        *        Request timed out.
 18 *        *        *        Request timed out.
 19 *        *        *        Request timed out.
 20 *        *        *        Request timed out.
 21 *        *        *        Request timed out.
 22 88 ms    88 ms    87 ms    google-public-dns-a.google.com [8.8.8.8]
Trace complete.

```

Source	Destination	Protocol	Len	Info
192.168.1.101	192.168.1.1	DNS	...	Standard query 0x8c5e PTR 8.8.8.8.in-addr.arpa
192.168.1.101	192.168.1.1	DNS	...	Standard query 0x8c5e PTR 8.8.8.8.in-addr.arpa
192.168.1.1	192.168.1.101	DNS	...	Standard query response 0x8c5e PTR 8.8.8.8.in-addr.arpa PTR goog.
192.168.1.1	192.168.1.101	DNS	...	Standard query response 0x8c5e PTR 8.8.8.8.in-addr.arpa PTR goog.
192.168.1.101	8.8.8.8	ICMP	...	Echo (ping) request id=0x0001, seq=206/52736, ttl=1 (no response)
192.168.1.1	192.168.1.101	ICMP	...	Time-to-live exceeded (Time to live exceeded in transit)
192.168.1.101	8.8.8.8	ICMP	...	Echo (ping) request id=0x0001, seq=207/52992, ttl=1 (no response)
192.168.1.1	192.168.1.101	ICMP	...	Time-to-live exceeded (Time to live exceeded in transit)
192.168.1.101	8.8.8.8	ICMP	...	Echo (ping) request id=0x0001, seq=208/53248, ttl=1 (no response)
192.168.1.1	192.168.1.101	ICMP	...	Time-to-live exceeded (Time to live exceeded in transit)
192.168.1.101	192.168.1.1	DNS	...	Standard query 0x247f PTR 1.1.168.192.in-addr.arpa

192.168.1.101	8.8.8.8	ICMP	...	Echo (ping) request	id=0x0001, seq=268/3073, ttl=21	(no respon
192.168.1.101	8.8.8.8	ICMP	...	Echo (ping) request	id=0x0001, seq=269/3329, ttl=22	(reply in
8.8.8.8	192.168.1.101	ICMP	...	Echo (ping) reply	id=0x0001, seq=269/3329, ttl=46	(request i
192.168.1.101	8.8.8.8	ICMP	...	Echo (ping) request	id=0x0001, seq=270/3585, ttl=22	(reply in
8.8.8.8	192.168.1.101	ICMP	...	Echo (ping) reply	id=0x0001, seq=270/3585, ttl=46	(request i
192.168.1.101	8.8.8.8	ICMP	...	Echo (ping) request	id=0x0001, seq=271/3841, ttl=22	(reply in
8.8.8.8	192.168.1.101	ICMP	...	Echo (ping) reply	id=0x0001, seq=271/3841, ttl=46	(request i